Tree Borrows

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Rust's type system provides powerful optimizations

```
fn write_both(x: &mut i32, y: &mut i32) -> i32 {
  *x = 13;
  *y = 20;
  *x
}
```

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```

mutable thus disjoint

*x has known value

*x is unchanged

always returns 13

Rust's type system provides powerful optimizations

```
fn write_both(x: &mut i32, y: &mut i32) -> i32 {
  *x = 13;
  *y = 20;
  13 // formerly *x: one fewer load from memory
}
```

Type-level guarantees for references

~~aliasing~~
~~&~~
~~mutability~~

`&mut` → mutation, no aliasing

`&` → aliasing, no mutation

Escape hatch: `unsafe`

Can **bypass typechecks** to implement **low-level manipulations**

```
unsafe {
  // Code within this block has relaxed typechecking
  ...
}
```

Within `unsafe` it is **the programmer's responsibility** to check

- that pointers are non-null
- that memory is initialized
- …

What if `unsafe` code is misused ?

```rust
fn write_both(x: &mut i32, y: &mut i32) -> i32 {
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}

fn main() {
  let mut root = 42;
  let ptr = &raw mut root;
  let x = unsafe { &mut *ptr };
  let y = unsafe { &mut *ptr };
  println!("{}", write_both(x, y));
}
```

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Not the compiler's responsibility

Within `unsafe` it is **the programmer's responsibility** to check

- that pointers are non-null
- that memory is initialized
- compliance with aliasing rules NEW!

Tree Borrows (TB): defines those aliasing rules

Not the compiler's responsibility

Within `unsafe` it is **the programmer's responsibility** to check

- that pointers are non-null
- that memory is initialized

Sounds familiar?

Stacked Borrows has the same purpose,

Tree Borrows is its successor.

Stacked Borrows (SB)

In safe Rust, the Borrow Checker makes borrows well-bracketed.
Stacked Borrows extends the well-bracketedness to unsafe.

```
let mut root = 42;
let ptr = &raw mut root;
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Desired outcome: UB

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```

root

- new stack at `root`

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ptr
root

- ✓ root is at the top
- push ptr

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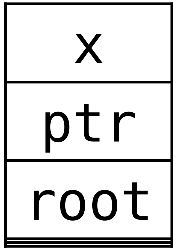

- ✓ `ptr` is at the top
- push `x`

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x
ptr
root

- pop until `ptr` is at the top

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y
ptr
root

- pop until `ptr` is at the top
- push y

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```

y
ptr
root

- search for `x`

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UB!

y
ptr
root

Can't use x if it is not in the stack

Desired outcome: UB

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$\rightarrow$ included in many projects' CI
$\rightarrow$ several bugs detected (e.g. in stdlib)

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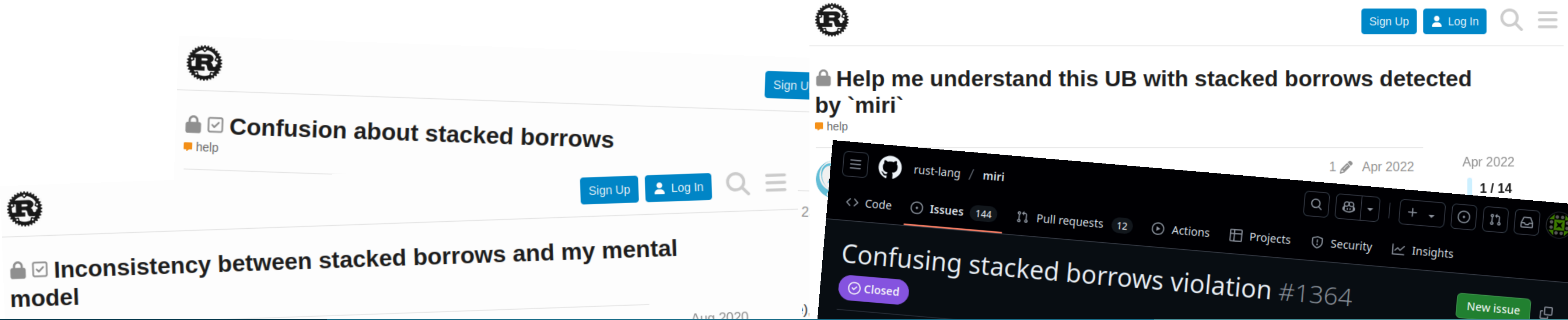

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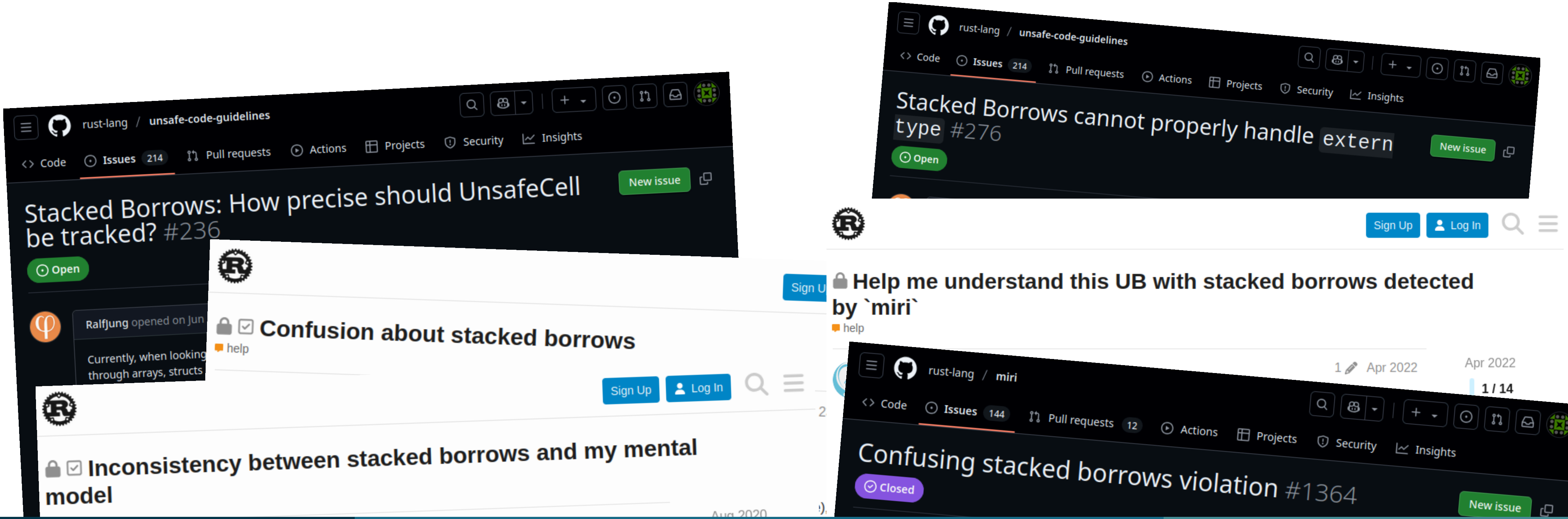
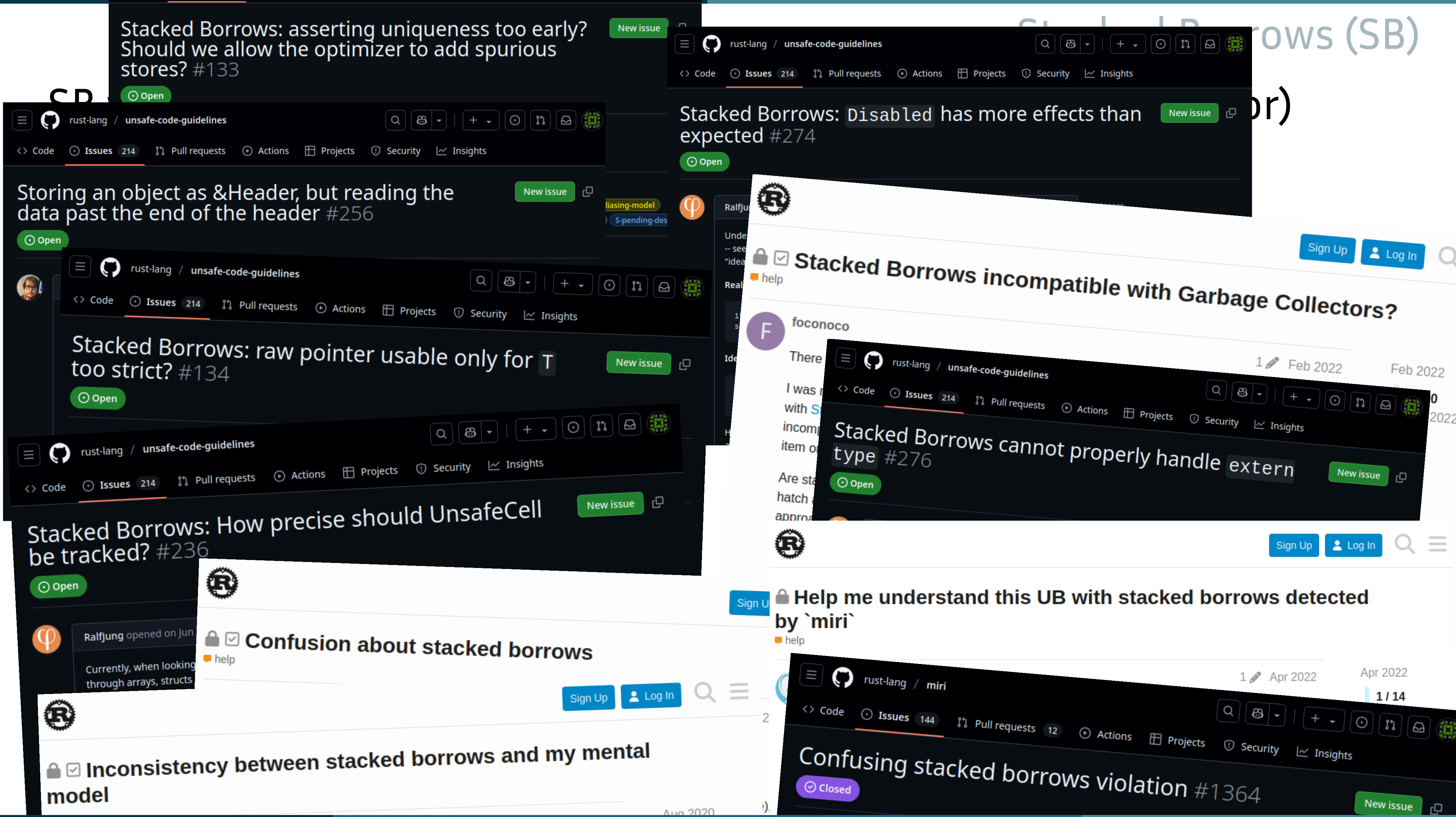
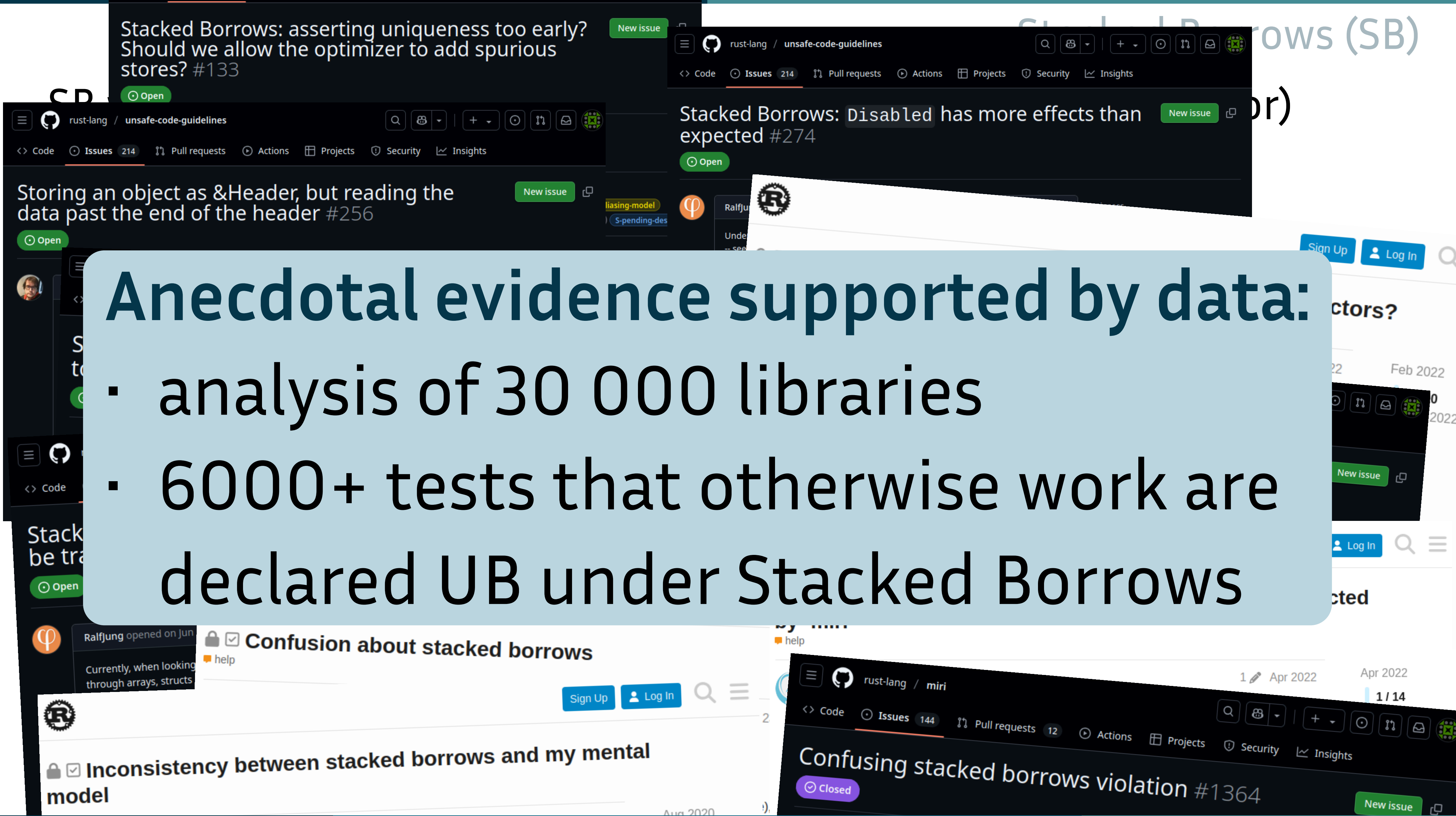




Stacked Borrows: asserting uniqueness too early? Should we allow the optimizer to add spurious stores? #133
Open
rust-lang / unsafe-code-guidelines
Stacked Borrows: Disabled has more effects than expected #274
Open
Storing an object as &Header, but reading the data past the end of the header #256
Open
Stacked Borrows: raw pointer usable only for T too strict? #134
Open
Stacked Borrows incompatible with Garbage Collectors?
help
foconoco
Stacked Borrows cannot properly handle extern type #276
Open
Stacked Borrows: How precise should UnsafeCell be tracked? #236
Open
Help me understand this UB with stacked borrows detected by `miri`
help
Confusion about stacked borrows
help
rust-lang / miri
Confusing stacked borrows violation #1364
Closed
Inconsistency between stacked borrows and my mental model

Anecdotal evidence supported by data:

- analysis of 30 000 libraries
- 6000+ tests that otherwise work are declared UB under Stacked Borrows



Tree Borrows allows much more code

Tree Borrows uses a **tree** instead of a stack to track borrows

Out of 30 000 most downloaded libraries,
$> 50\%$ **fewer** tests with aliasing UB when using Tree Borrows

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fixes known technical limitations of SB, incl. **pointer offsets**

From Stacks to Trees

```
let mut root = vec![0, 0, 0];
let x0 = &raw mut root[0];
let x2 = &raw mut root[2];
unsafe { *x0 = 1 };
unsafe { *x2 = 3 };
```

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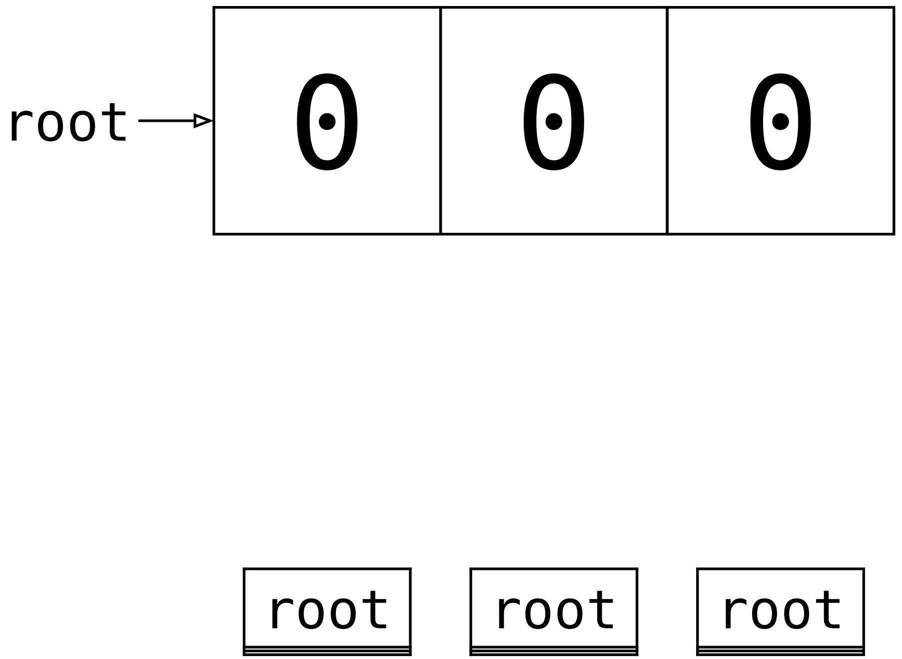

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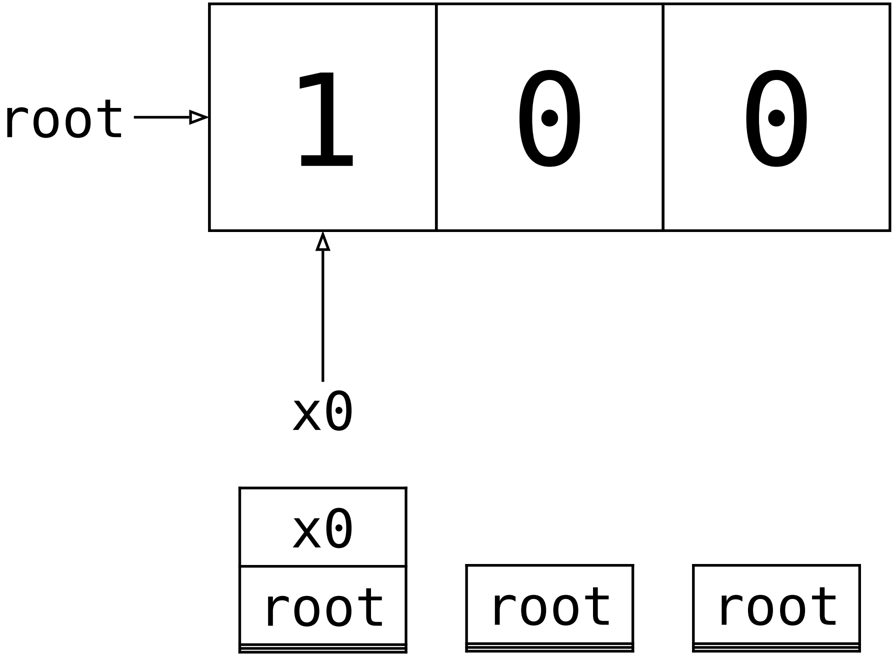

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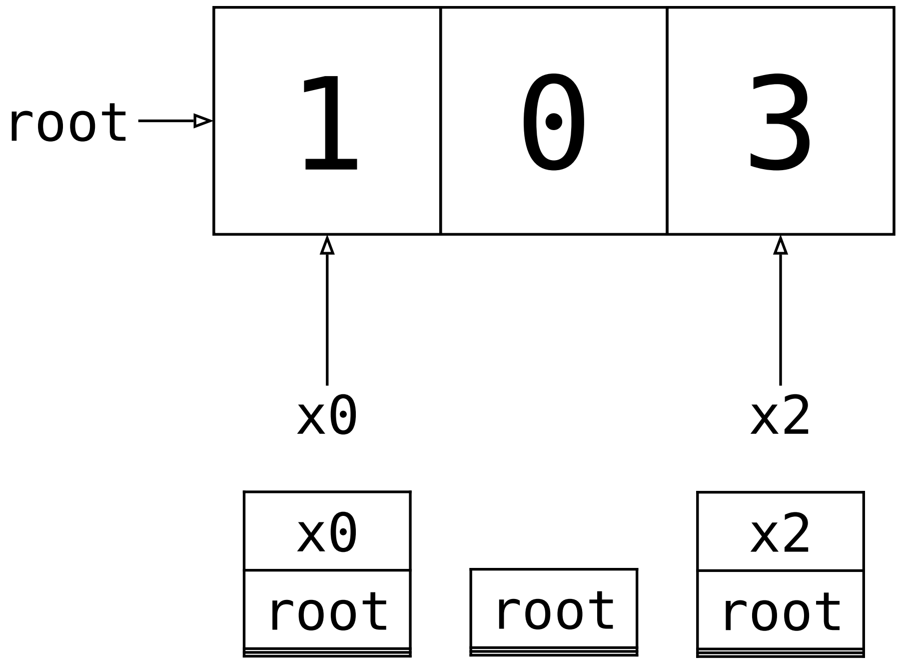

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unsafe { *x0.add(1) = 2; }
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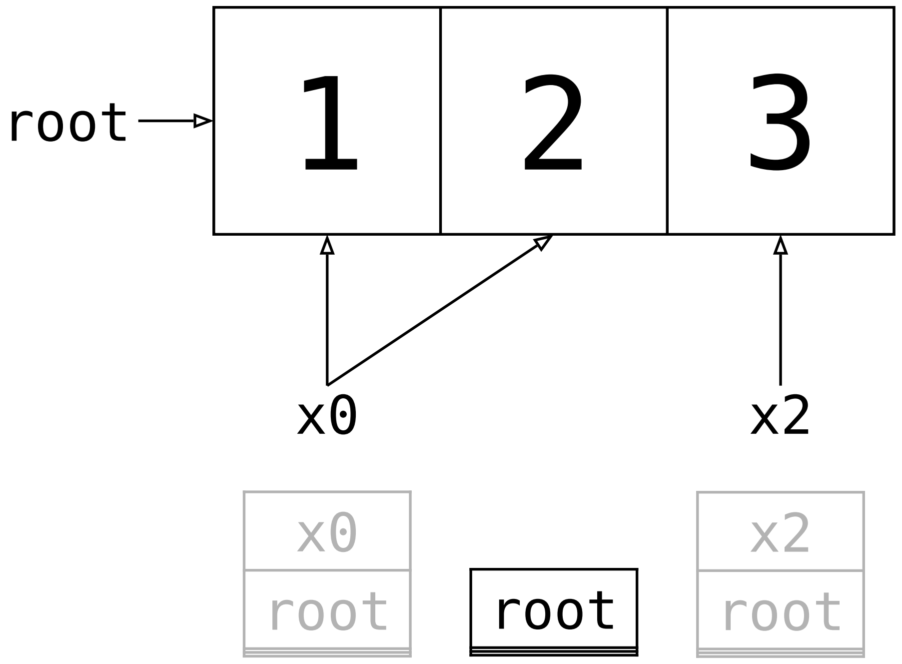

Desired outcome: not UB

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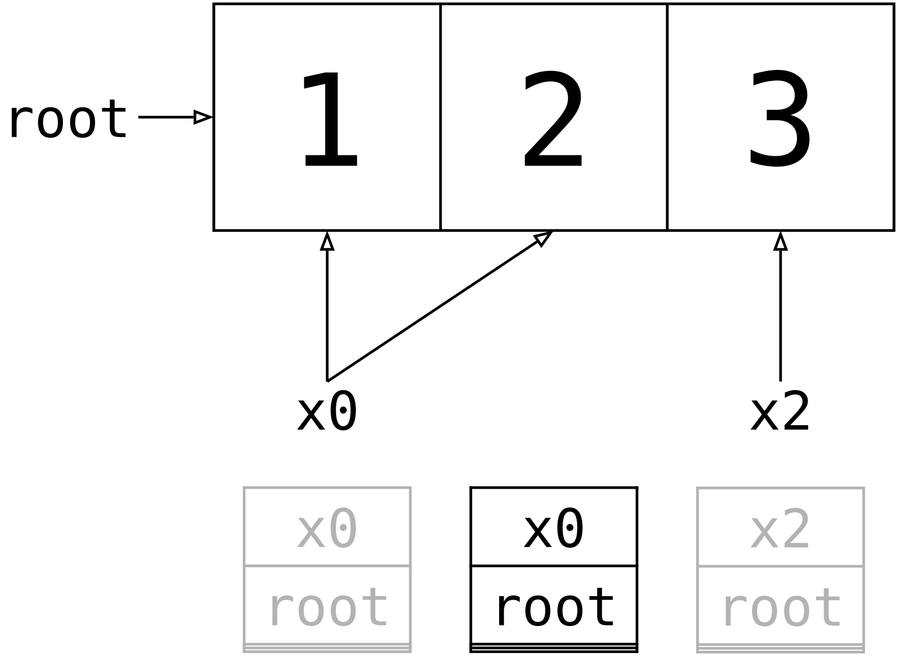

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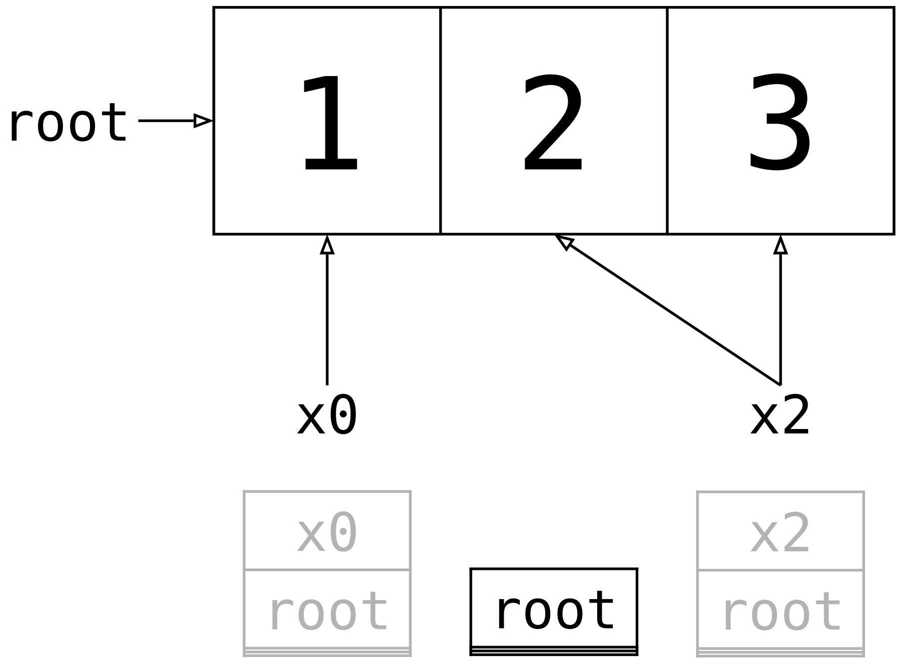

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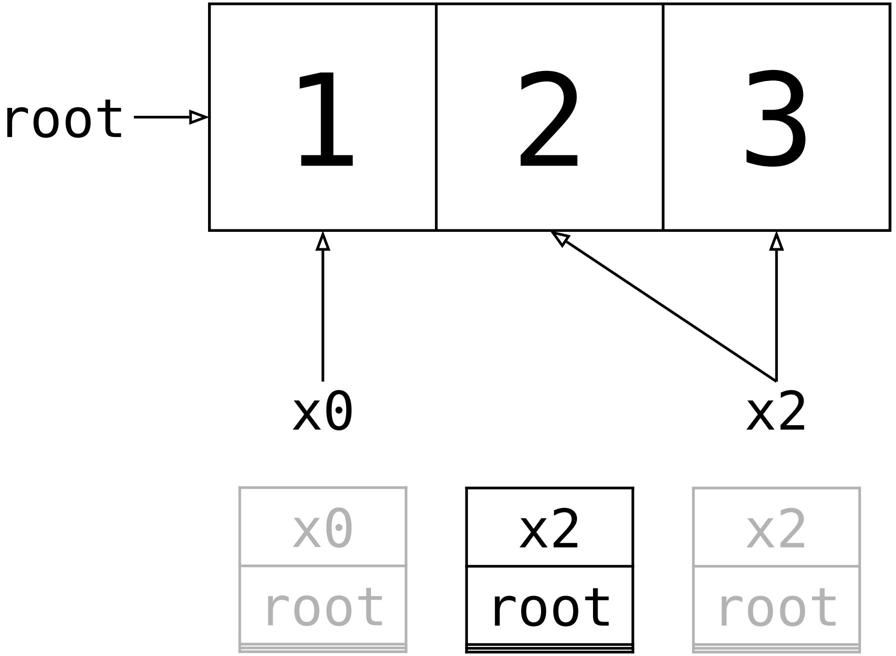

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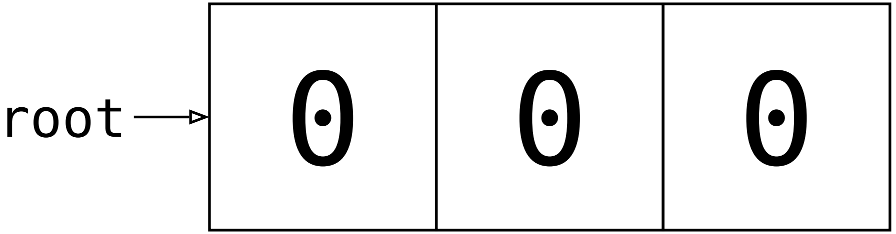

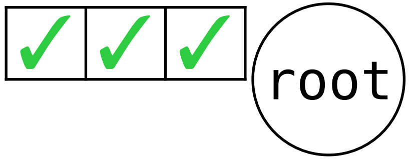

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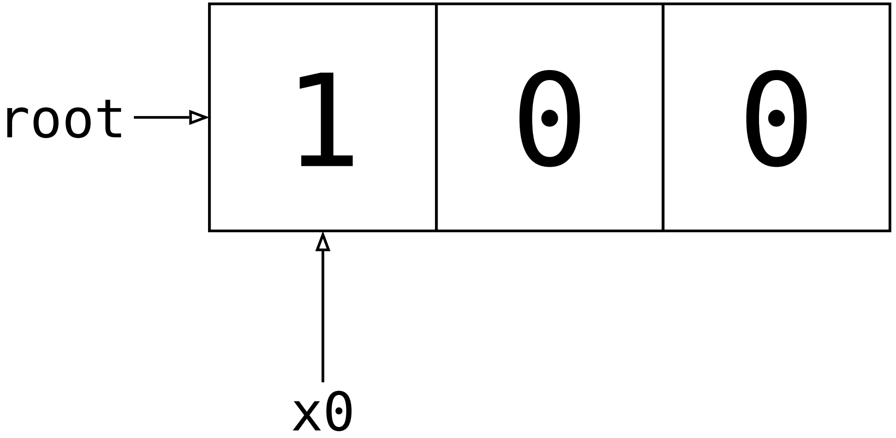

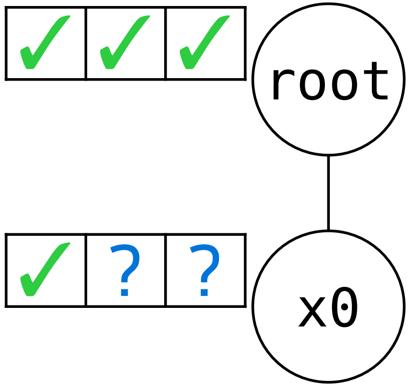

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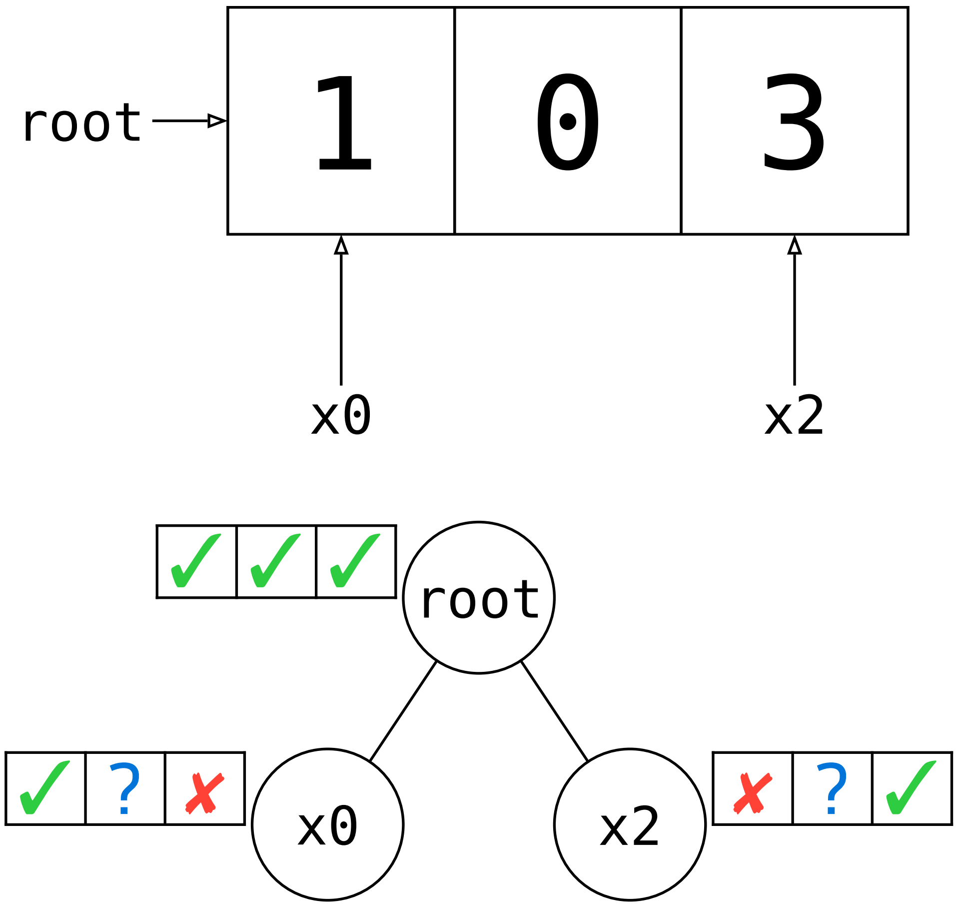

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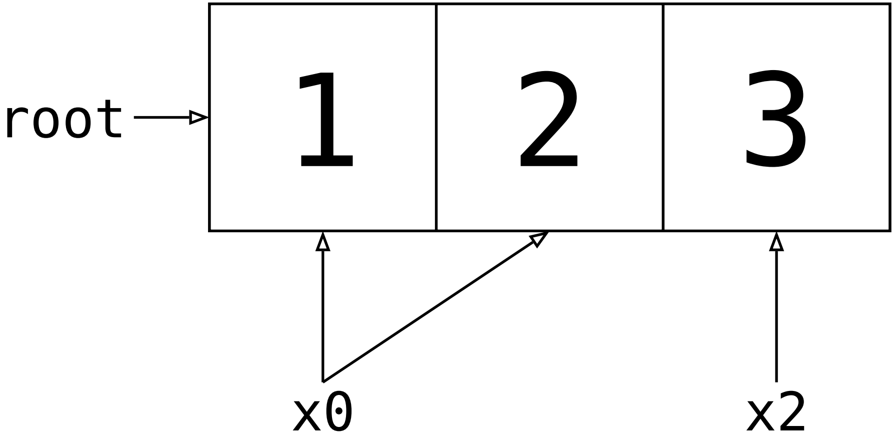

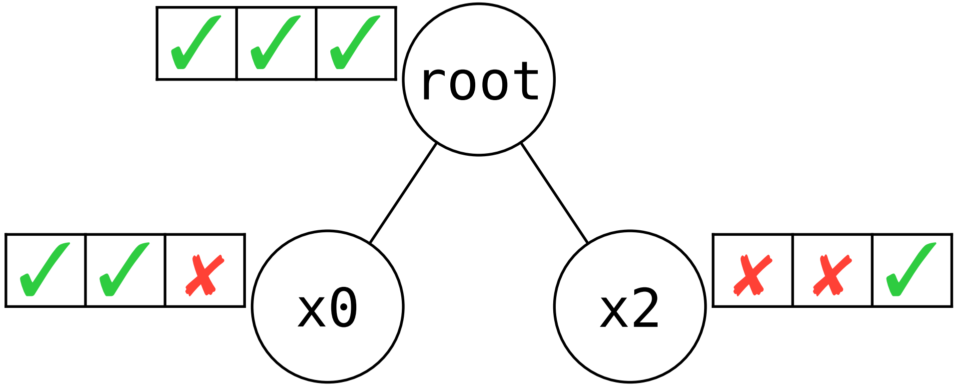

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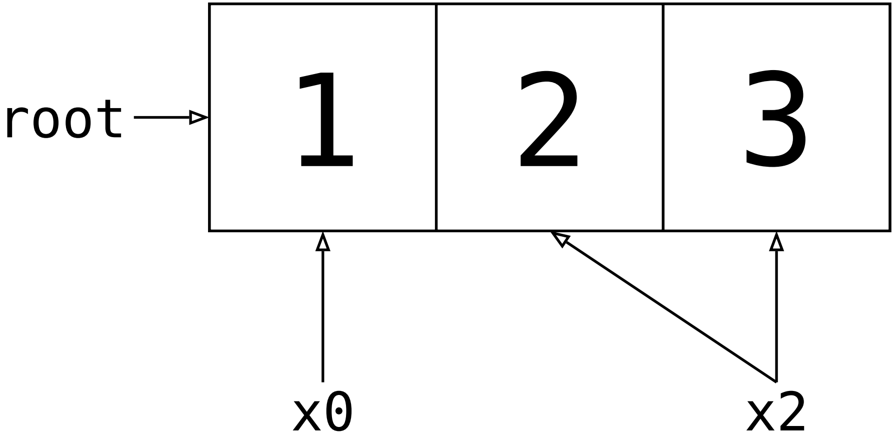

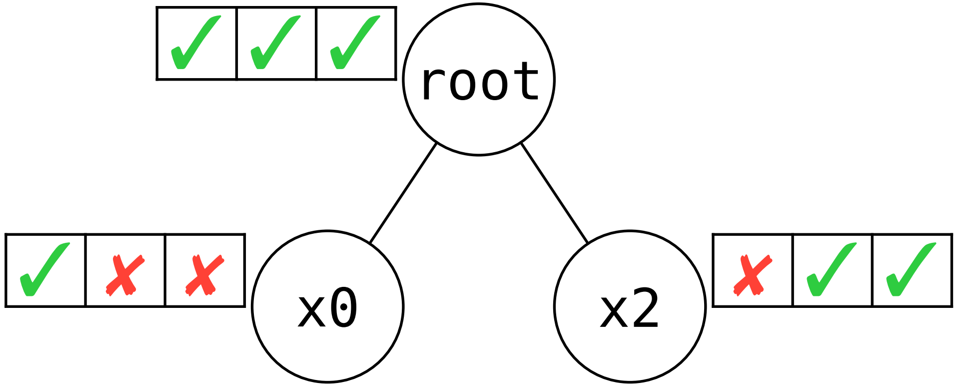

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A second look at the motivating example


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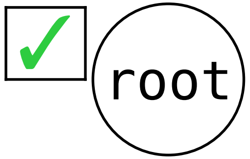

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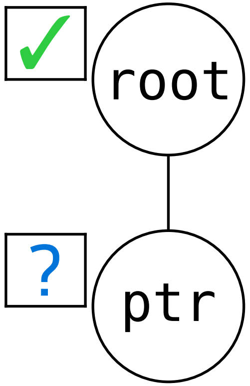

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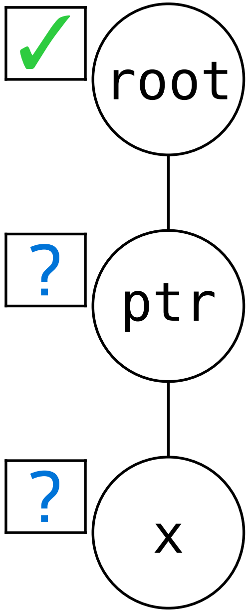

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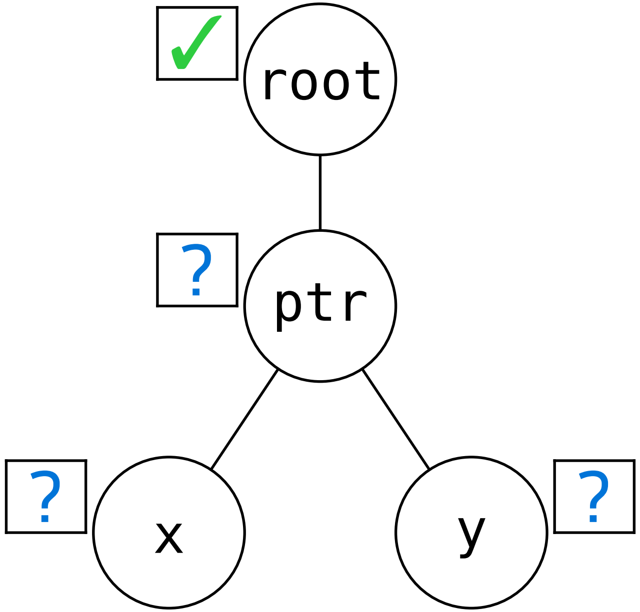

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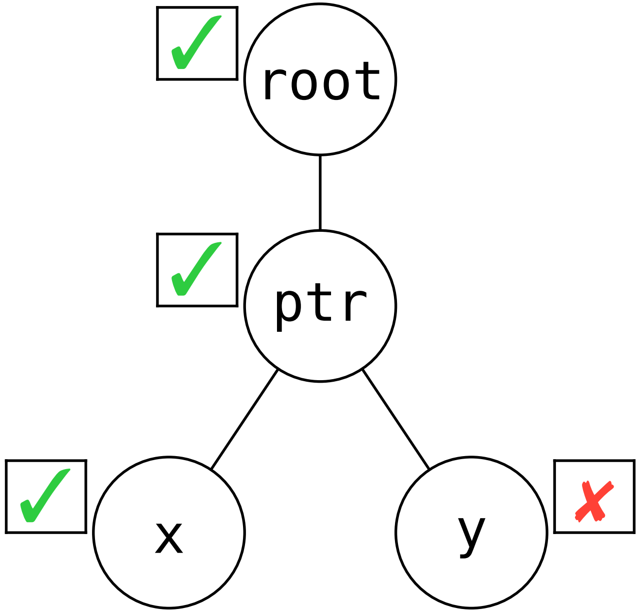

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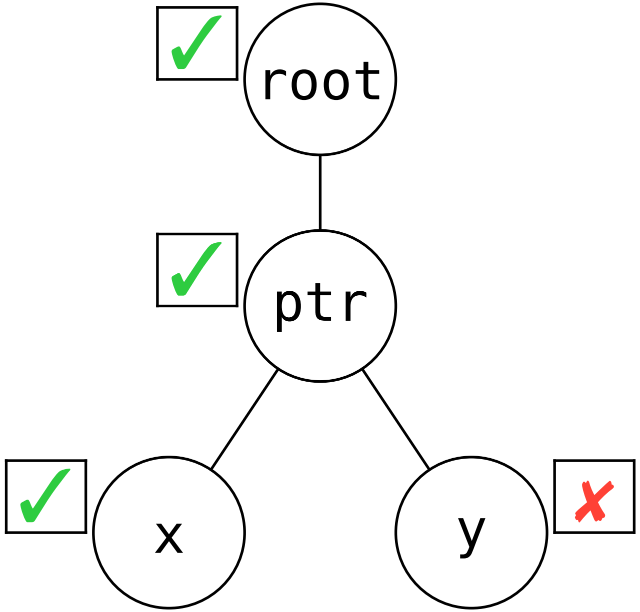

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Evaluation

TB should enable desired optimizations

i.e. have enough UB to rule out problematic patterns

- formalized in Rocq (+Simuliris)
- a selection of optimizations proven
  - ✓ delete read through `&mut` or `&`
  - ✓ insert read through `&` in function
  - ✓ move read down for `&mut` or `&` in function

  ...

It should be possible to write unsafe code free of UB

i.e. UB should be predictable and not too common

Tree Borrows reduces aliasing-related UB by over 50%

Only 31 ($< 0.01\%$) tests are regressions, all easily fixable.

(Out of 30 000 libraries, 400 000+ working tests)

“Tree Borrows accepts more real-world programs that call foreign functions than Stacked Borrows due to differences in handling pointer arithmetic.”

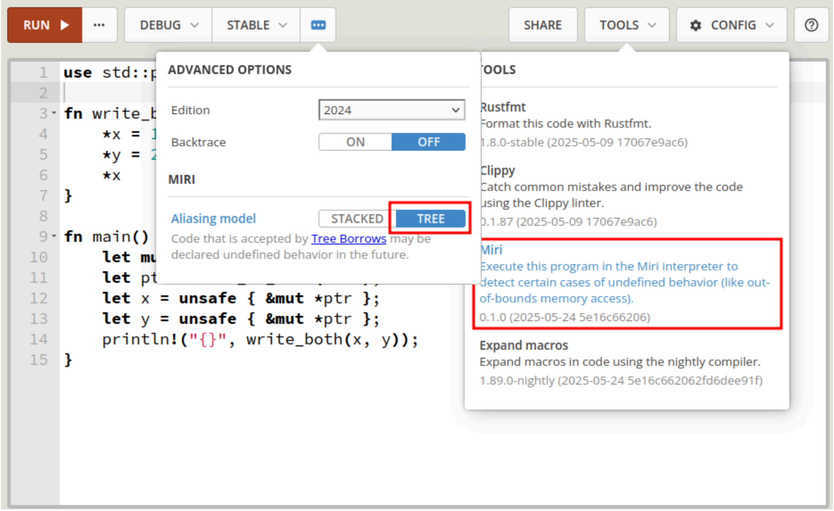
A Study of Undefined Behavior Across Foreign Function Boundaries in Rust Libraries, by I. McCormack, J. Sunshine, J. Aldrich @ ICSE’25

Conclusion

Try it out:

Rust Playground supports TB

`play.rust-lang.org`



Learn more:

`plf.inf.ethz.ch/research/pldi25-tree-borrows.html`

- detailed state machine
- raw pointers
- interior mutability